ZCAS UNIVERSITY

SCHOOL OF COMPUTING, TECHNOLOGY AND APPLIED SCIENCES

COMPUTER SCIENCE DEPARTMENT

END OF SEMESTER EXAMINATIONS

COURSE: OBJECT ORIENTED PROGRAMMING WITH JAVA
COURSE CODE: CCS 2121

TIME ALLOWED: 3 Hours
TOTAL MARKS: 100 Marks

GENERAL INSTRUCTIONS TO CANDIDATES:

1. This paper contains **FIVE (5)** questions, each carrying **25 marks.**
2. Answer **ANY FOUR (4)** questions.
3. Each question has sub-questions labelled (a) to (e), each carrying **5 marks.**
4. **Neat and clear work** is highly recommended.
5. Write your student number clearly on all answer sheets.
6. Begin each question on a new page.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

QUESTION 1

a) Java is described as an architecture-neutral and platform-independent programming language. Explain what this means, referring to the role of the Java Virtual Machine (JVM) and bytecode in enabling Java programs to run on different operating systems. ***[5 marks]***

b) Explain the key structural elements of a Java program by describing the purpose of each of the following: (i) comments, (ii) reserved words, (iii) modifiers, (iv) blocks, and (v) the main method. Use appropriate code snippets to support your answer. ***[5 marks]***

c) A student wrote the following Java program but it contains errors. Identify and correct ALL errors, explaining clearly why each is an error: public Class HelloWorld { public static void Main(String args) { system.out.println("Hello World") } } ***[5 marks]***

d) Discuss THREE types of programming errors that can occur in Java programs: syntax errors, runtime errors, and logic errors. ***[5 marks]***

e) Describe the complete process of creating, compiling, and running a Java application from the command line. ***[5 marks]***

QUESTION 2

a) Java supports eight primitive data types. List and describe any FIVE of them, specifying for each: the data type name, its size in bits, the range of values it can hold, and an appropriate use case. Provide a declaration and initialization example for each. ***[5 marks]***

b) Explain the difference between implicit (widening) casting and explicit (narrowing) casting in Java. Given the following declarations: ***byte b = 50; int i = 1000; double d = 9.99***; State clearly, with justification, which of the following assignments are valid, which require explicit casting, and which will cause a compilation error:

(i) int x = b; (ii) byte y = i; (iii) int z = d; (iv) double w = i; (v) float f = 1.4; ***[5 marks]***

c) Evaluate and explain the output of the following Java code segment step by step: int a = 0; int x = a + (++a); int b = 5; int y = b++ * 2; int c = 10; int z = (++c) + c; System.out.println("x=" + x + " y=" + y + " z=" + z); Explain the concepts of prefix and suffix increment operators and operand evaluation order in Java. ***[5 marks]***

d) Java uses operator precedence and associativity to determine how expressions are evaluated. Given the expression: int i = 3; int result = 3 + 4 * 4 > 5 * (4 + 3) - ++i ? 10 : 20; Show the full step-by-step evaluation, identifying the order in which each operator is applied and the final value stored in result. ***[5 marks]***

e) Write a complete Java program called TemperatureConverter that (i) declares a double constant for the conversion formula factor (5.0/9.0), (ii) declares a double variable for a Fahrenheit temperature of 98.6, (iii) computes the equivalent Celsius

temperature using the formula C = (F - 32) * (5/9), and (iv) prints both values formatted clearly. Also identify and correct the common integer division pitfall in this formula. ***[5 marks]***

QUESTION 3

a) Java provides both if-else and switch statements for branching. Write a Java program that takes an integer score (0-100) as input and uses a nested if-else if-else structure to print the corresponding grade: A (90-100), B (80-89), C (70-79), D (60-69), and F (below 60), along with a descriptive remark for each grade. Explain why a switch statement would not be ideal for this particular scenario. ***[5 marks]***

b) Explain the boolean operators &&, ||, !, and ^ (exclusive OR) in Java. For each operator, write a meaningful Java code example that demonstrates a real-world use case (e.g., validating user input, checking access rights. ***[5 marks]***

c) A student is applying for a university scholarship. The criteria are: (i) GPA must be 3.5 or above, (ii) the student must not have any disciplinary record, and (iii) the student must be either enrolled in a Science programme OR have community service hours greater than 100. Write a complete Java program using boolean variables and compound boolean expressions to determine if a student qualifies. Test your program with two different sets of values. ***[5 marks]***

d) Explain what a ternary (conditional) operator is in Java. Write a Java program that uses the ternary operator to: (i) determine if a number is even or odd, (ii) find the maximum of three numbers, and (iii) determine if a year (e.g., 2024) is a leap year (divisible by 4 but not 100, or divisible by 400). Show equivalent if-else versions for comparison. ***[5 marks]***

e) Consider the following program requirements: a simple Java calculator that reads two double values and an operator (+, -, *, /) from the user and performs the corresponding operation. Write the complete Java program using a switch statement to handle each operation. Include appropriate error handling for (i) division by zero and (ii) an invalid operator symbol entered by the user. ***[5 marks]***

QUESTION 4

a) A university wants to model a Student entity. Design a Student class with the following: (i) at least four appropriate instance variables with correct access modifiers, (ii) a no-argument constructor, (iii) a parameterized constructor, and (iv) appropriate getter and setter methods. ***[5 marks]***

b) Explain the following key object-oriented concepts as they apply in Java, providing a short code example for each: (i) Inheritance create a Person superclass and a Student subclass that extends it, (ii) Method Overriding override a displayInfo() method in the subclass, (iii) The super keyword demonstrate its use to call the superclass constructor,

(iv) Polymorphism show how a Person reference variable can hold a Student object, and (v) the instanceof operator. ***[5 marks]***

c) Write a Java class BankAccount that uses: (i) a static variable to count the total number of accounts created, (ii) instance variables for accountNumber and balance, (iii) a static method to display the total count, and (iv) instance methods for deposit, withdraw, and getBalance. Demonstrate its use in a main method. ***[5 marks]***

d) Write a complete Java class Rectangle with (i) a no-arg constructor that sets length and width to 1.0, (ii) a constructor that takes only one parameter (for a square), and (iii) a constructor that takes both length and width. Include a computeArea() method and demonstrate all three constructors in a main method. ***[5 marks]***

e) Explain what the ***this*** keyword means in Java and describe THREE different ways it can be used: (i) to refer to instance variables when there is shadowing by local variables, (ii) to call another constructor in the same class (constructor chaining), and (iii) to pass the current object as an argument to another method. ***[5 marks]***

QUESTION 5

a) Distinguish clearly between: (i) a void method and a value-returning method, (ii) formal parameters and actual arguments, and (iii) pass-by-value in Java. ***[5 marks]***

b) Write a method computeCircleArea(double radius) and a method swap(int a, int b), then demonstrate that Java does not support pass-by-reference for primitives. ***[5 marks]***

c) Write a complete Java program that: (i) declares and initialises an integer array of 10 elements with values {5, 3, 8, 1, 9, 2, 7, 4, 6, 10}, (ii) implements a method to compute and return the sum and average of the elements ***[5 marks].***

d) Recursion is a technique where a method calls itself. Explain the two essential components of any recursive method: the base case and the recursive case. Write recursive Java methods for: (i) computing the factorial of a non-negative integer n, ***[5 marks]***

e) Scope and lifetime of variables are critical concepts in Java. Explain the difference between local variables, instance variables, and class (static) variables in terms of: (i) where they are declared, (ii) their scope (where they can be accessed) ***[5 marks]***

END OF EXAMINATION PAPER

"Integrity in examination is your responsibility. Any form of academic dishonesty is strictly prohibited."